CONVERSIONS

G	x10^9
M	x1000000, x10^6
K	x1000, x10^3
	x1, x10^0
c	x0.01, x10^-2
m	x0.001, x10^-3
μ	x10^-6
n	x10^-9
p	x10^-12
f	x10^-15

V	Voltage	Volts
I	Current	Amps
R ρ	Resistance	Ohms
T	Time	Second
P	Power	watts
Q	Charge	Coulombs
F	Capacitance	Farads
W	width	m
L	length	m
⍺	activity factor	unit

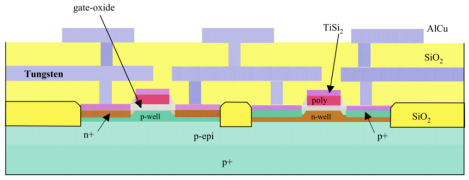
BASIC FORMULAE

V=IR
Voltage = Current * Resistance
(volts = amps*ohms)

P=IV
Power = Current * Voltage
(watts = amps * volts)

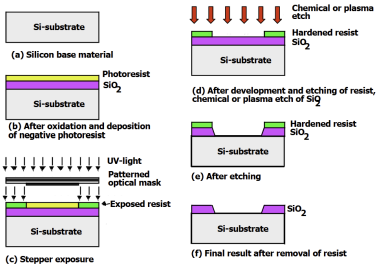
LECTURE 1 INTRO

Moore's law: number of transistors
on chip doubles every 2 years.
Typical CMOS structure (image below)



Dual-Well Trench-Isolated CMOS Process

CMOS fabrication steps:
• Transfer an image of the layer design to wafer
• Do something to the imaged part of the wafer
• Implant, Deposit, Grow, Etch, etc
• Strip off image, sometimes polish wafer to make it flat
Entire die image transferred at once
Entire wafer surface processed at once
Economical because everything done in parallel



Fundamental Constraints of Fabrication Layout:
Resolution (Smallest width feature,
smallest spacing without short circuits)
Alignment/overlap (Layers have to be aligned,
Tolerance required to be manufactured)
Density rules, Cost, Area, Power usage, etc.

Design rules: constraints from fabrication standards.
- minimum poly width/spacing
- minimum metal width/spacing
- minimum active region width/spacing
- contact region size/spacing

Integrated Circuit Costs:
Fixed/Non-recurring, independent of volume made.
Example: design time/effort, mask generation, equipment
Recurring/Variable cost, dependent on volume made.
Example: silicon processing, packaging, testing.
Fixed Costs increasing with every new tech generation.

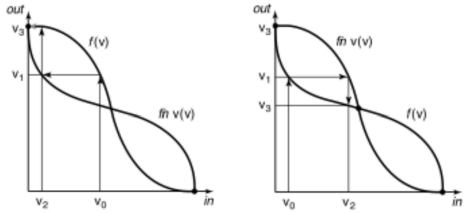
Device in Circuit in Gate in Module in System.

Packaging requirements:
Electrical: low parasitics.
Mechanical: reliable and robust
Thermal: Good heat removal
Economical: cheap!

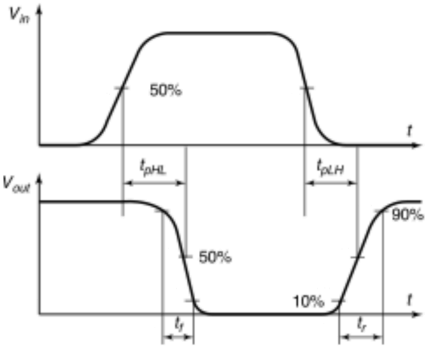
Bonding of chip to board:
Wire bonding
Tape automated bonding
Flip Chip bonding
Package to Board interconnect
-through holes or surface mount

LECTURE 2 INVERTER DESIGN/ANALYSIS

Regenerative property:
Even if input signal/voltage forced into noise region,
distributed signal converges back into nominal range.
Example: chain of inverters.
Below: regenerative and non regenerative voltages



Delay definitions:



LECTURE 3 DELAY OPTIMIZATION

Instantaneous power: P=Vt+It
P watts, V Voltage, I current, t time.

Peak power: P_{peak}=Vdd * I_{peak}
Watts = Source voltage* peak current

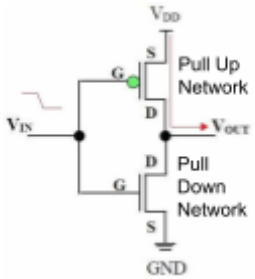
Average power: Vdd/tau * t->t+τ∫I(t) dt
watts=source voltage/time period, *
(current*time) integrated from t to t+tau.

Energy efficient metric = energy*delay.

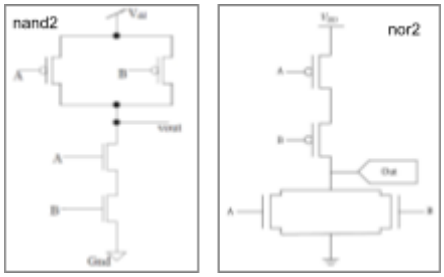
Energy from power source:
$P=C_L Vdd^2$
power = Load Capacitance* Source Voltage^2
W = F*V^2

Energy Stored in Capacitor:
P=½ * C*Vdd^2
power = 0.5*Capacitance*Source Voltage^2
W =constant*F*V^2

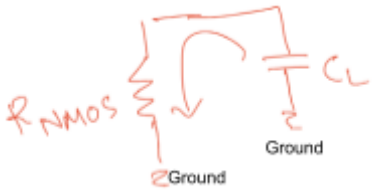
Static Complementary CMOS logic
Pull up network: PMOS.
Pull Down Network: NMOS.
PUN and PDN are dual logic networks.
PUN and PDN functions are complementary.



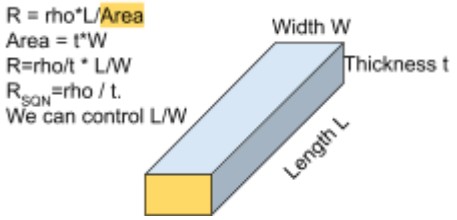
SCL Static CMOS Logic properties
Full rail-to-rail swing at output
Symmetric VTC (Voltage Transfer Characteristic)
No Static Power Dissipation
Propagation Delay function of transistor capacitance & resistance



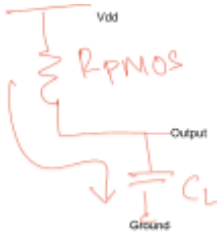
Inverter Delay Optimization:
Design the circuit for driving output capacitance (C_L)
Resistors can be inverters.



$R_{NMOS} = R_{SQN} * (L/W_{NMOS})$
R_{NMOS}: resistance of NMOS.
R_{SQN}: Resistance of material - decided in manufacturing.
L: Length of transistor, W: Width of transistor.



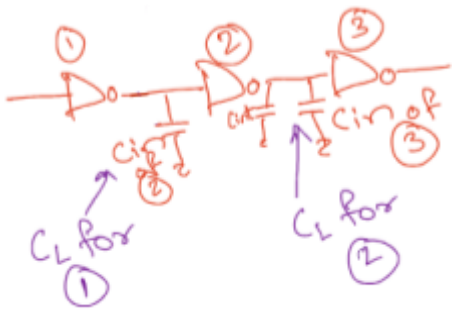
Tau_{HL} = time delay from output change to high to low output change
Tau_{HL}= R_{NMOS} * C_L (time = resistance * farads) *ln(2) if Vdd->0.5Vdd
High to low delay = Nmos resistance * Load capacitance



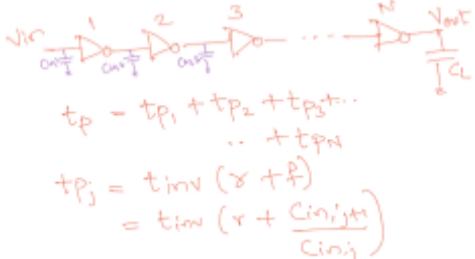
R_{PMOS}=R_{SQP} * L/W_P
Tau_{LH} = Time delay low to high
Tau_{LH} = R_{PMOS}*C_L
(*ln(2) if 0->0.5Vdd)
R_{SQP} = rho/thickness

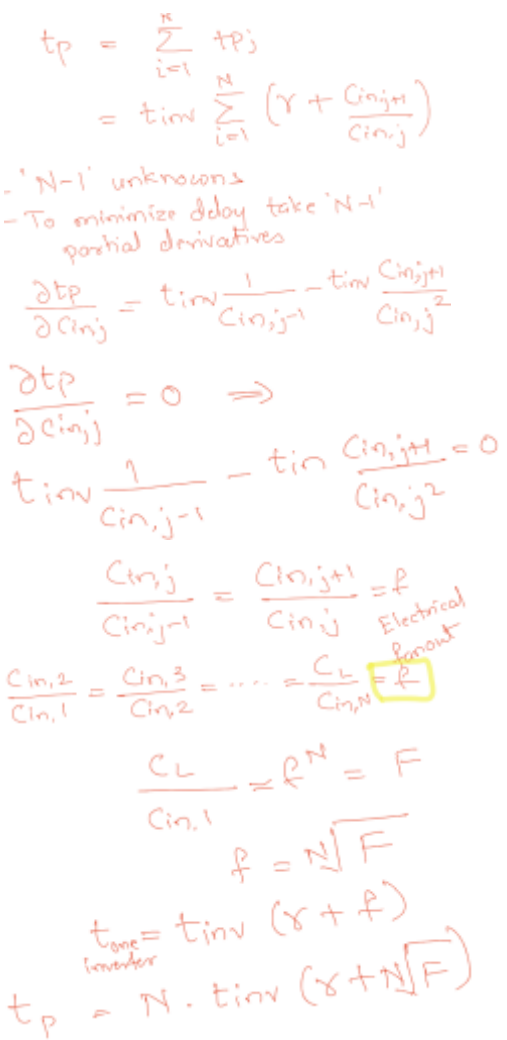
Remember:
Tau_{HL} = Tau_{LH}
R_{NMOS} = R_{PMOS}
R_{SQN}: L/W_N = R_{SQP}: L/W_P
Rule of thumb: $2W_N$ = W_P

C_{INT} = Internal capacitance. Also known as Diffusion Capacitance.
C_L = Load capacitance = Input capacitance of downstream gate.
C_{IN} = Input capacitance = C_L of upstream gate.
C_{IN} = C_G (W_P + W_N). if $2W_N$ = W_P, C_{IN} = $3WC_G$.
C_{INT} = Internal Capacitance. C_{INT} = $C_D(W_P+W_N)$.
C_G decided in the tech department not by designers.



ɣGamma = C_{INT}/C_{IN} = C_D/C_G. should be 1 to 1.5. Unaffected by L, W
Electrical Fanout f = C_L/C_{IN}
Tau_{HL} = ln2 * R_{NMOS}* C_{IN}(ɣ+f) = $t_{INVERTED}$(ɣ+f).
$t_{INVERTED}$ NOT DELAY OF INVERTER; IS DELAY OF OUTPUT FLIP!



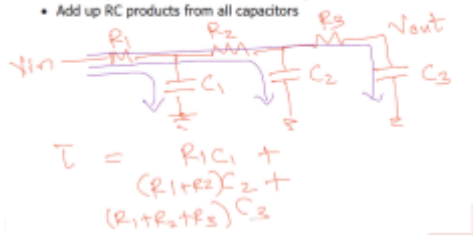


Optimum electrical fanout f: 3.6 for ɣ = 1, 4 for ɣ=1.5.
t_p N t_{INV}(ɣ+N*sqrt(F)). F = C_L/C_{IN}.
$C_{TOT} = C_{IN}+fC_{IN}+F^2 * C_{IN} +...+ f^N *C_{IN}$
$C_{TOT} = C_{IN}(1+f^N)$ + overhead
overhead = $C_{IN}(f+f^2+...+f^{N-1})$

LECTURE 4 GATE DELAY & LOGICAL EFFORT

Elmore delay: only count resistors along paths!

- For each capacitor draw path of current from capacitor to input
- Multiply each capacitance value with the sum of resistances on the current path that are common with the path from Vin to Vout
- Add up RC products from all capacitors



Logical effort: gates

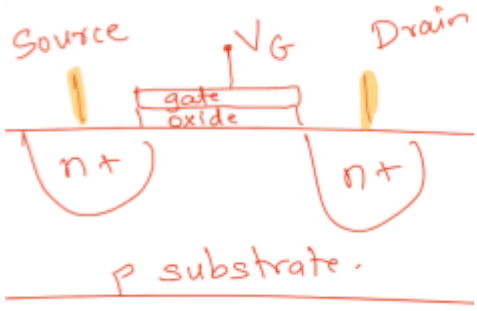
Gate type	Logical Effort for different number of inputs					
	1	2	3	4	5	n
Inv	1					
NAND		4/3	5/3	6/3	7/3	(n+2)/3
NOR		5/3	7/3	9/3	11/3	(2n+1)/3
Mux		2	2	2	2	2
XOR,XNOR		2	6	16		

Gate type	Parasitic delay
Inv	p_{inv} = 1
n-input NAND	$n\,p_{inv}$
n-input NOR	$n\,p_{inv}$
n-way Mux	$2n\,p_{inv}$
n-input XOR,XNOR	$4n\,p_{inv}$

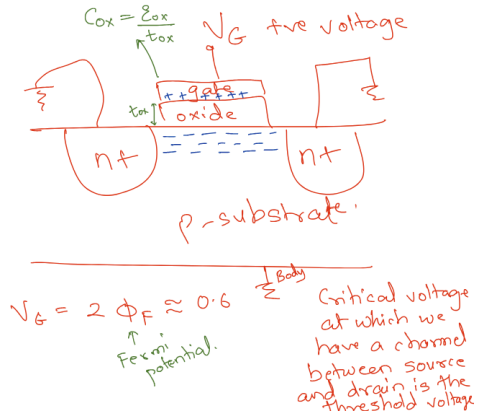
Logical effort(LE) of gate i = LE_i
Stage effort: $h_i = LE_i f_i$
Path logical effort: $G = LE_1LE_2...LE_N$
Path electrical effort: $F = C_{OUT}/C_{IN}$
Branching effort: $B = b_1b_2...b_N$
Path effort: H = GFB
Path delay $D = \Sigma d_i = \Sigma p_i + \Sigma h_i$
Find the best number of stages: N ~ $\log_4 H$
Compute the stage effort: $h = H^{1/N}$
Sketch path with this number of stages
Work from either end, find sizes:
$C_{IN} = C_{OUT}*LE/h$

LECTURE 5 MOS TRANSISTOR MODEL

Capacitance from Gate to Oxide = C_{OX} * W*L

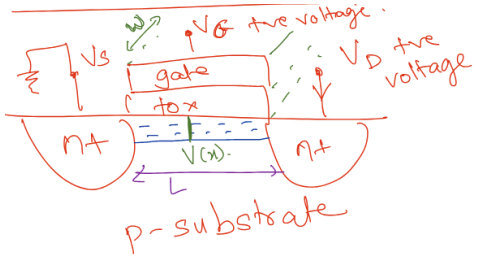


Case 1: Source and Drain and Bulk/Body are grounded, positive energy applied to gate. V_T = Threshold Voltage, ɣ = Body Effect Coefficient.
$V_T = V_{T0}+ɣ(\sqrt{(|-2\Phi_F+V_{SB}|)}-\sqrt{(|2\Phi_F|)})$



$V_G = 2\Phi_F \approx 0.6$ (Fermi potential.)
Σ^{Body} Critical voltage at which we have a channel between source and drain, is the threshold voltage

Case 2: Applying Drain Bias

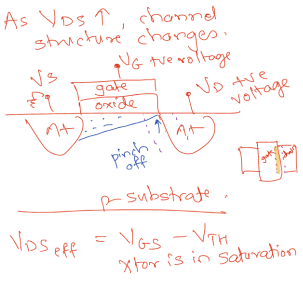


$I_D = Q(x) v_n$ W ← width of the xtor.
Q: charge density. v_n: drift velocity of the carriers

Q=CV(charge=capacitance*voltage)=$C_{OX}(V_{GS}-V(x)-V_{TH})$

$v_n(x) = \mu E$ (mobility, electric field)

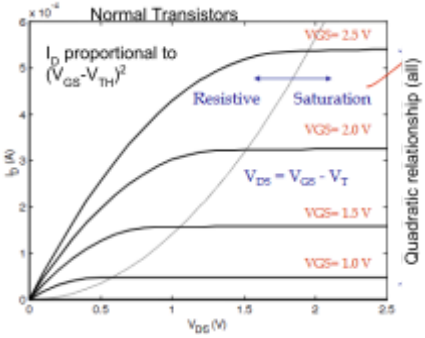
I_D = (W/L) (mobility) (C_{OX}) $[(V_{DS}(V_{GS}-V_{TH}) - V^2_{DS}/2]$
Mobility * C_{OX} = k'_n = process transconductance.
k'_n * W/L = Gain Factor.

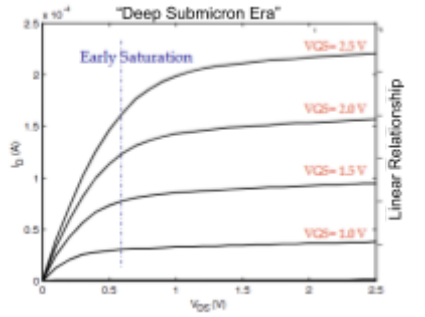


$V_{DS\,eff} = V_{GS} - V_{TH}$ xtor is in Saturation

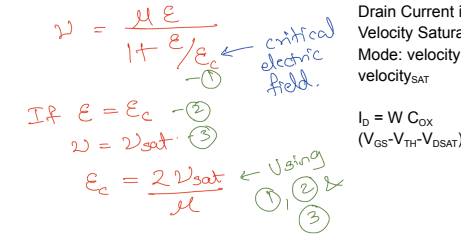
Case 3: As V_{DS} increases, channel structure changes - Pinch Off.
$I_D = k'_n/2$ * W/L * $(V_{GS} - V_{TH})^2$
More precisely: Multiply by $(1+\lambda V_{DS})$.
λ = Channel Length Modulation Parameter.

Modes of Operation:
Cutoff. $V_{GS} - V_{TH} < 0$; $I_D = 0$.
Linear (resistive): $V_{GS} - V_{TH} >= 0$; $V_{GS} - V_{TH} > V_{DS}$.
$I_D = k'_n$ * W/L * $[(V_{GS} - V_{TH})*V_{DS}]$
$R_{eff} = 1 / [k'_n$ * W/L * $(V_{GS} - V_{TH})]$
Saturation: $V_{GS} - V_{TH} >= 0$; $V_{GS} - V_{TH} < V_{DS}$.
$I_D = k'_n/2$ * W/L * $(V_{GS} - V_{TH})^2$ * $(1+\lambda V_{DS})$.



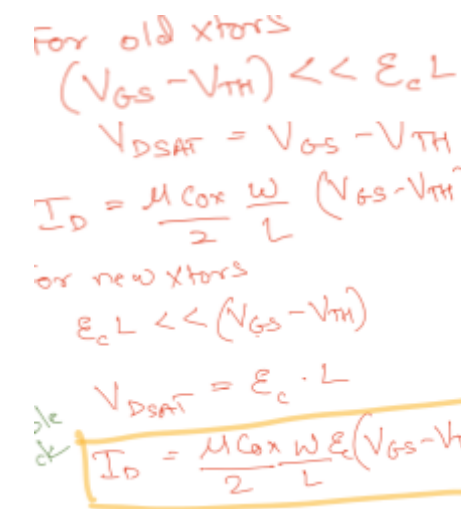


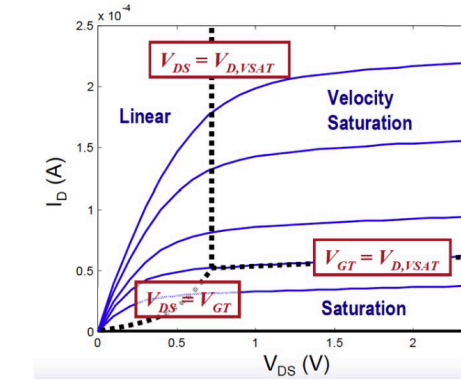
When Long Channel, V_{GS} graphed against I_D quadratic whole time.
When Short channel, V_{GS} graphed against I_D quadratic then linear.



Drain Current in Velocity Saturation Mode: velocity in $velocity_{SAT}$

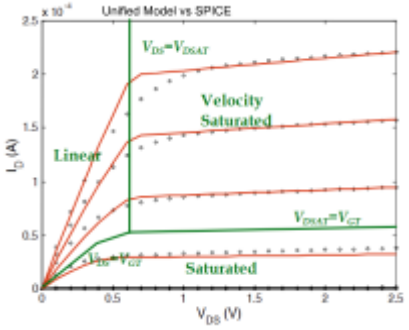
I_D = W C_{OX} $(V_{GS}-V_{TH}-V_{DSAT})veloc_{sat}$

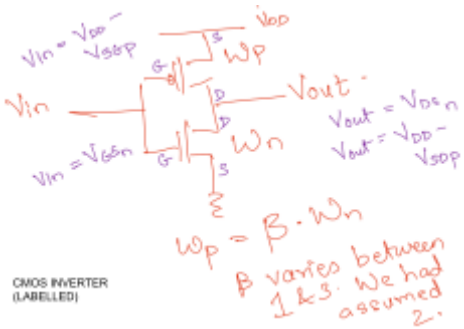




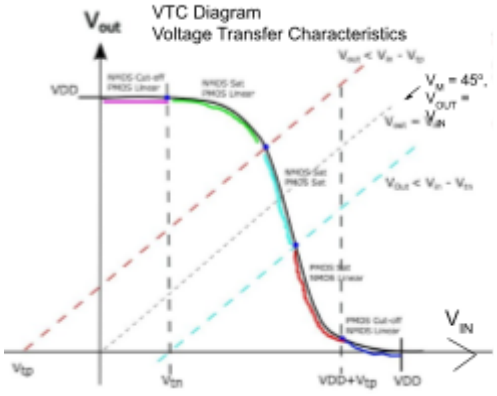
Transistor Models:
1: Velocity Saturation Model (discussed earlier in doc)
2: alpha-power law.∝ = Alpha - activity factor?
I_D proportional to $(V_{GS} - V_{TH})^{\alpha}$
3: V_T^* Model -> always assume velocity saturated.
$I_D = k'_n$ * W/L * $[V_{GS} - V_{TH} - V_{DSAT}/2]V_{DSAT}$
$V_{DSAT}/2$ is $\acute{\varepsilon}_c L$ at velocity saturation.

$V_T^* = V_{TH} + V_{DSAT}/2$
$I_D = k'_n * W/L * [V_{GS} - V_T^*]V_{DSAT}$
4: Unified Model for Manual Analysis
$V_{GT} = V_{GS} - V_{TH}$
For $V_{GT} < 0$, $I_D = 0$.
For $V_{GT} >= 0$, $I_D = k'_n * W/L * [V_{GT} * V_{DS\,EFF} - V_{DS\,EFF}^2/2] (1+\lambda V_{DS})$
with $V_{DS\,EFF}$ = minimum (V_{GT}, V_{DS}, $V_{D\,vSAT}$).



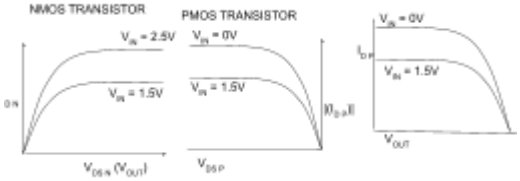


LECTURE 6 INVERTER VTC, MOS CAPACITANCE



	NMOS	PMOS
Operating regions		
	$V_{GSn} < V_{tn}$ (Cut-off)	$V_{GSp} > V_{tp}$ (Cut-off)
	$V_{GSn} > V_{tn}$	$V_{GSp} < V_{tp}$
	$V_{DSn} < V_{GSn} - V_{tn}$ (Linear)	$V_{DSp} > V_{GSp} - V_{tp}$ (Linear)
	$V_{DSn} > V_{GSn} - V_{tn}$ (Sat.)	$V_{DSp} < V_{GSp} - V_{tp}$ (Sat.)
Terminal voltages		
	$V_{GSn} = V_{in}$	$V_{GSp} = V_{in}$-VDD
	$V_{DSn} = V_{out}$	$V_{DSp} = V_{out}$-VDD

DC operation of inverter: $I_{D\,NMOS} = I_{D\,PMOS}$.

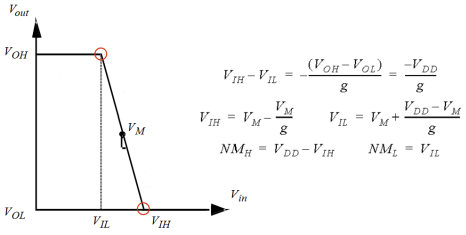


$V_{SD\,P} = -V_{DS\,P}$. $V_{OUT} = V_{DD} - V_{SD\,P}$. $V_{OUT} = V_{DD} + V_{DS\,P}$
(See inverter CMOS diagram in Lecture 3 section for D and S locations)
Graphically, we are interested in overlap points between NMOS graph and V_{OUT} I_{DP} graph when overlapped.
Here we use the V_{TH}^* Model.
$I_D = k'_n * W/L * [V_{GS} - V_T^*]V_{DSAT}$

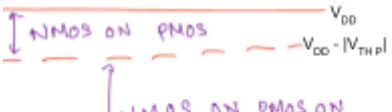
To find V_M: (below/next column)

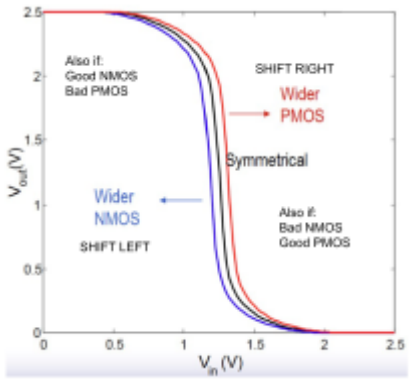
$$V_M = \frac{\left(V_{Tn} + \frac{V_{DSATn}}{2}\right) + r\left(V_{DD} + V_{Tp} + \frac{V_{DSATp}}{2}\right)}{1+r} \quad \text{with } r = \frac{k_p V_{DSATp}}{k_n V_{DSATn}} = \frac{\upsilon_{satp} W_p}{\upsilon_{satn} W_n}$$

Finding V_{IH} and V_{IL} :

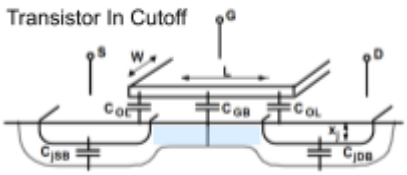


$$V_{IH} - V_{IL} = -\frac{(V_{OH} - V_{OL})}{g} = \frac{-V_{DD}}{g}$$
$$V_{IH} = V_M - \frac{V_M}{g} \qquad V_{IL} = V_M + \frac{V_{DD} - V_M}{g}$$
$$NM_H = V_{DD} - V_{IH} \qquad NM_L = V_{IL}$$





Capacitance from Gate to Oxide = C_{OX} * W*L
Channel has no terminal - lump part of the capacitance to S and D and B terminals. $C_{GS} = C_{GCS} + C_{GSO}$
C_{GCS} = Channel Capacitance Component
C_{GSO} = Overlap Capacitance Component



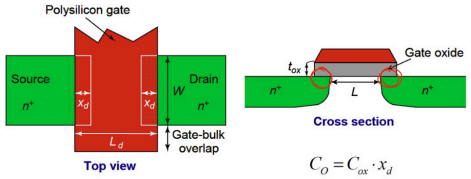
No carriers in the channel to form the other terminal of the capacitor
- Substrate acts as the other terminal
- Capacitance becomes a series combination of gate oxide and depletion capacitance

$|V_{GS}| < |V_T|$, total C_{GCB} is smaller than W*L*C_{ox} (approx $C_{GCB} = 0$)
In cutoff state, V_{GS} approaches zero. $C_{Depletion}$ increases.

In linear state, channel of charge forms. $C_{GCB} = 0$.
Changing either voltage changes Channel Charge.
Resistance equally distributed across the channel.

In saturation region, C_{GCS} about 2/3 * W * L * C_{OX}
Drain Voltage no longer affects channel charge - it's set by source and $V_{DS\,SAT}$. Resistance is unevenly distributed to one side.

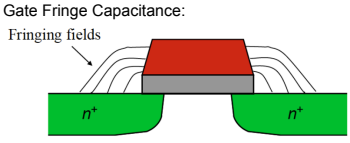
Gate Overlap Capacitance: when gate and oxide overlap with the N+ materials, causing leakage. $C_O = C_{OX} * X_D$ where X_D is the overlapped thickness of the transistor.



$$C_O = C_{ox} \cdot x_d$$

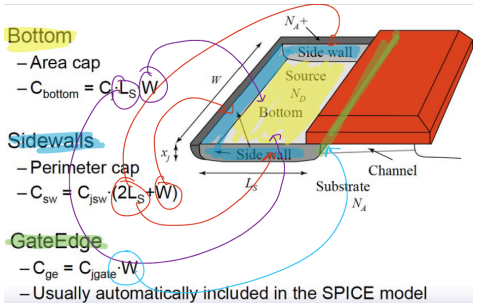
Off/Linear/Saturated ---> $C_{GSO} = C_{GDO} = C_O * W$

Gate Fringe Capacitance:



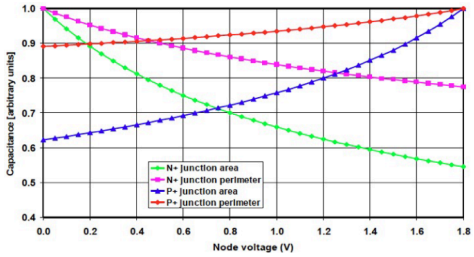
C_{OV} not just from metal overlap - comes from Fringing Fields too.
Typical Value: 0.2 fF (femtofarads) (capacitance)

Diffusion Capacitance



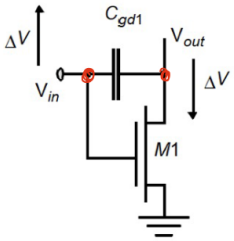
Junction Capacitance:
Junction Caps non-linear. C_J is a function of junction bias.
SPICE MODEL EQUATIONS:

$$\text{Area } C_J = \text{area} \times C_{J0} / (1 + |V_{DB}|/\phi_B)^{mj}$$
$$\text{Perimeter } C_J = \text{perim} \times C_{JSW} / (1 + |V_{DB}|/\phi_B)^{mjsw}$$
$$\text{Gate edge } C_J = W \times C_{Jgate} / (1 + |V_{DB}|/\phi_B)^{mjswg}$$



Miller Effect:
As V_{IN} increases, V_{OUT} drops. (Once in the transition region, the difference needed to get from V_{in} to V_{OUT} is <1.

So, C_{GD} experiences voltage swing greater than V_{IN}. Means you need to provide more charge. Makes C_{GD} look bigger than it is.

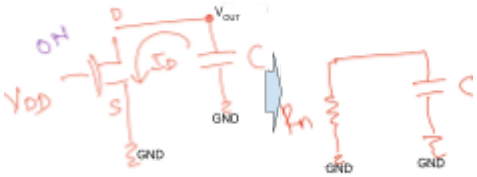




IN: 0 → V_{DD}. OUT: V_{DD} → 0.
Effective change in voltage is 2V_{DD}. C_{EFF} = 2C_{GD}.

LECTURE 7 CMOS DELAY

CMOS Switching Delay:



$I_D = C (dV_{DS} / dt)$: capacitance * difference voltage/difference time
$tp = ln(2) * R_N * C$

Which I_D equation do we choose to find R_N?
$V_{DD} \rightarrow V_{DD}/2$ uses Saturation. as $V_{DS} = V_{DD}$
In saturation, I_D proportional to $(V_{GS}-V_{TH})^Z$ (Z depends on long channel or short channel device)

Voltage will discharge linearly.
I = C (difference V/difference time)
Difference time = $(V_{DD} - V_{DD}^2)*C/I_{D\,SAT}$
Note: $I_{D\,SAT} = I_{D\,SAT\,0}(1+\lambda V_{DS})$
$V_{OUT} = (V_{DD} + 1/\lambda)e^{(-t)/(C/\lambda\ VDS)} - 1/\lambda$
For small λ, $tp = (C * V_{DD}/2)/((+\lambda V_{DD}) * I_{D\,SAT\,0})$

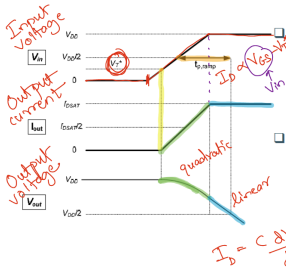
$R_N = V_{DD}/(2ln(2) * I_{D\,SAT\,0})$
2ln(2) approx 0.72

$t_{pHL} = ln(2)\ C_L\ R_{eqn}$. $t_{pLH} = ln(2)\ C_L\ R_{eqp}$.
$R_{eq} = V_{DD}/ (2\ ln(2)(1 + \lambda V_{DD})\ I_{D\,SAT})$

Ideal scenario for output change: straight lines, no curves, no delay
Real life: output change not instant, even S-curved!

$I_{PMOS} - I_{NMOS} = C_L * dV_{OUT}/dt$
$I_{PMOS} = V_{GSpmos} * t$, $I_{NMOS} = V_{GS\,nmos} * t$

Graphical analysis of slope dependence (1)

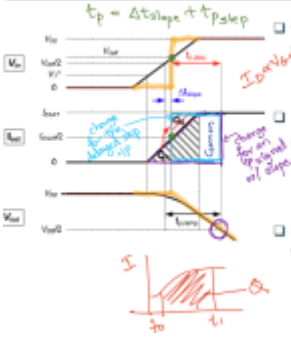


- I_{out} ramps linearly for $V_T^* < V_{in} < V_{DD}$
  - Constant once $V_{in} = V_{DD}$
- C_L integrates I_{out}
  - $V_T^* < V_{in} < V_{DD}$: V_{out} quadratic
  - $V_{in} = V_{DD}$: V_{out} linear

$I_D = C\frac{dV}{dt}$

Graphical analysis of slope dependence (2)

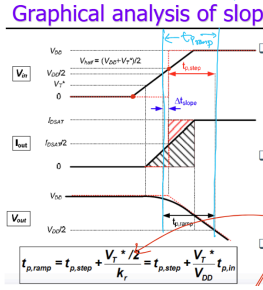
$t_p = \Delta t_{slope} + t_{pstep}$



- Consider step input whose output crosses $V_{DD}/2$ at same time
- V_{out} set by charge removed from C_L
  - Need to make $Q_R = Q_S$
- Step has to shift to when $I_{out} = I_{DSAT}/2$

$t_{pramp} = t_{pstep} + \Delta t_{slope}$

Graphical analysis of slope dependence (3)



- To find Δt_{slope}:
  - Find V_{in} when $I_{out} = I_{DSAT}/2$ (V_{half})
  - And use input t_r
- I_{DSAT} α $(V_{DD}-V_T^*)$:
  $V_{half} - V_T^* = V_{DD}/2 - V_T^*/2$
  $V_{half} = (V_{DD}+V_T^*)/2$
- So $\Delta t_{slope} = (V_T^*/2)/k_r$
  - $k_r = V_{DD}/(2*t_{p,in})$

$$t_{p,ramp} = t_{p,step} + \frac{V_T^*/2}{k_r} = t_{p,step} + \frac{V_T^*}{V_{DD}} t_{p,in}$$

Find difference in t slope:
$I_{D\,SAT} * V_{HALF} = I_{D\,SAT} * V_{DD}/2$
$I_{D\,SAT}$ proportional to $(V_{GS} - V_T^*)$
$V_{HALF} = (V_{DD}+V_T^*)/2$
Slope of Input Signal:
$k_r = V_{DD}/(2*t_{p\,in})$

For reasonable Input Slopes,
$t_{p\,ramp} = t_{p\,step} + (V_T^*/V_{DD}) * t_{p\,in}$
For $t_{p\,average}$, V_T^* is $(V_{TN}^* + V_{TP}^*)/2$
V_{TN}^* / V_{DD} typically about 1/3 at nominal supply (standard voltage)

Fanout is the same across a chain of inverters.
$t_{p1} = t_{p\,step}$, $t_{p2} = t_{p\,step} + V_T^*/V_{DD} * t_{p1}$

For Large N (number of inverters) where $V_T^*/V_{DD} < 1$

$$t_{pk} = t_{pstep}\left(\sum_{i=0}^{k-1}\left(\frac{V_T^*}{V_{DD}}\right)^i\right)$$

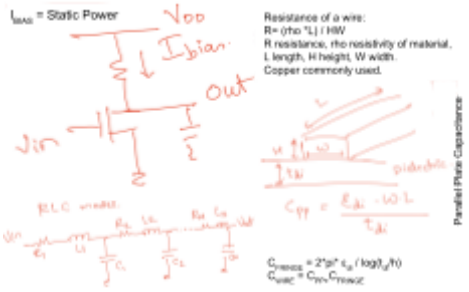
$$t_{pk} = \frac{\ln(2)\,RC}{1 - \frac{V_T^*}{V_{DD}}}$$

LECTURE 8 CMOS POWER, WIRES
CMOS POWER DISSIPATION:
Dynamic Power
$E_{0\to1} = C_L * V_{DD}^2$, $E_C = 0.5 * C_L * V_{DD}^2$
Power = Energy* Transition * transition rate
transition rate = transitions per second
Power (E) = $C_L * V_{DD}^2 * f_{0\to1}$
Power (E) = $C_L * V_{DD}^2 * f * P_{0\to1}$
(where $P_{0\to1}$ = Probability of a 0→1 transition)
$E = C_{SWITCHED} * VDD * f$ (where $C_{SWITCHED} = C_L * P_{0\to1}$)

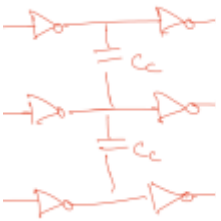
Transition Activity & Power
Energy consumed in N cycles.
$E_N = C_L * V_{DD}^2 * N_{0\to1}$
$N_{0\to1}$ = number of 0→1 transitions in N cycles

$P_{AVERAGE}$ = limit as N→infinity $(N_{0\to1}/N) = \propto_{0\to1}$
$\propto_{0\to1}$= Activity Factor for 0→1 transitions
$P_{AVERAGE} = \propto_{0\to1} * C_L * V_{DD} * f$

Static Power:



Coupling Capacitance:

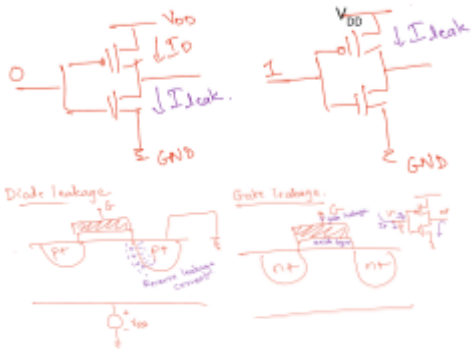


All wires switching in same direction at same time: $C_{EFF} = 0$.

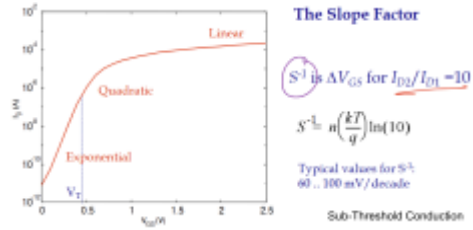
Neighbouring wires switching in opposite direction: $C_{EFF} = 2C_C$

Neighbouring wires are grounded: $C_{EFF} = C_C$

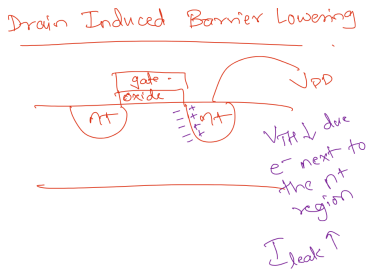
Leakage Power:



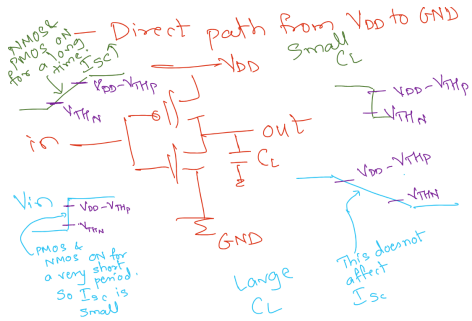
Transistor Leakage
$I_D = I_0 e^{(q(VGS-VTH)/nkT)}$
I_0: nominal current
n (Sub Threshold slope factor) = $1+C_D/C_{OX}$
C_D = depletion capacitance



If VGS < VTHN: in sub threshold regime.
$I_D = I_0 e^{(q(VGS-VTHN)/nkT)} * (1-e^{q* VDS/kT})$



Short-circuit Power



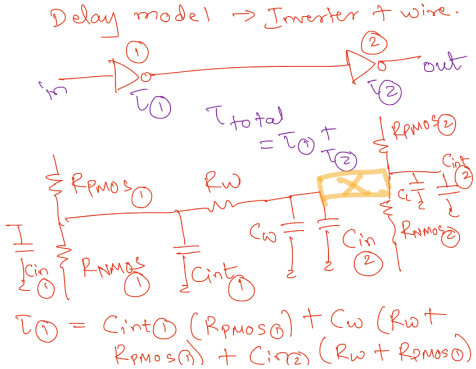
Slew rate/Rise time/Fall time are important. Affects performance and short circuit power.

RC delays should only be considered when t_{pRC} is much smaller than $t_{p\,gate}$ of the driving gate. Lcrit way bigger than sqrt($t_{p\,gate}$/0.38rc)

RC delays should only be considered when the rise (fall) time at the wire input is smaller than the rise (fall) time of the wire. $t_{rise} < RC$
• when not met, the change in the signal is slower than the propagation delay of the wire.

Wire model: lumped model, where resistance of the wire is in one resistor R_W. $Tau_{LUMPED} = R_W C_W$.
Distributed Model, where each wire segment is represented by a register. ($Tau_{DISTRIBUTED} = C_{SEGMENT1}*(R_{SEGMENT1}) + C_{SEGMENT2}*(R_{SEGMENT1} + R_{SEGMENT2}) + C_{SEGMENT3}*(R_{SEGMENT1} + R_{SEGMENT2} + R_{SEGMENT3})$ etc.)
→Simplifies to $rC_L^2/N^2 * N(N+1)/2$.
If N is large, $Tau_{DISTRIBUTED} = rC_L^2/2 = R_W C_W/2$

$R_{SEGMENT1} = R_{SEGMENT1} = R_{SEGMENT1} = (r *$ length of segment)
r= material resistance/length
$C_{SEGMENT1} = C_{SEGMENT1} = C_{SEGMENT1} = (c *$ length of segment)
c= material capacitance/length
delta L = L wire length/N number of segments.

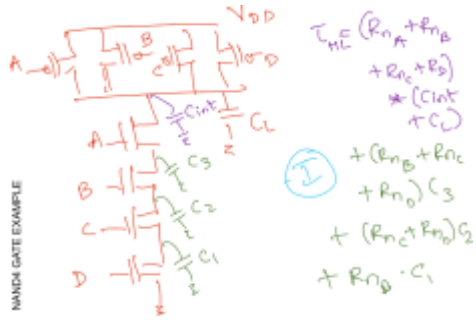


$Tau_2 = C_{INT(2)} * R_{NMOS2} = C_L * R_{NMOS2}$
$Tau_{TOTAL} = Tau_1 + Tau_2$

Wire Stack in VLSI systems - Stacked in layers above transistors, 10 to 12 metal layers in current CMOS tech, orthogonal routing in alternate layers to reduce coupling capacitance.

Local interconnect: close, thin wires connecting very close transistors.
Global Interconnect: large, thick wires connecting one area of chip to another.

LECTURE 9 CMOS LOGIC
see lecture 3 for pull up/pull down network and examples



Propagation Delay (t_p) deteriorates rapidly as a function of Fan In. Gates with a fan-in greater than 4 should be avoided.

Reducing Parasitic Delay (tau):
1: Transistor sizing. W up, R down, C up. Delay proportional to RC.
$Tau = R_N * C_{INTERNAL} + R_N * C_L$.
First term doesn't change, Secondterm decreases.
Don't think of gates in isolation: think of entire combinational path.
2: Progressive Transistor Sizing.
$W_{N\,D} > W_{N\,C} > W_{N\,B} > W_{N\,A}$ (from nand gate example)
3: Input Reordering. if an input is the last to switch, position it so that the rest of the capacitances from the other transistors can drain to ground without it.
4: Reduce Voltage Swing. $Tau_{P\,HL} = 0.5 * C_L * V_{SWING} / I_{SAT\,N}$.
instead of 0→V_{DD}, try 0→ V_{DD}/x.

ADDITIONAL REFERENCES
Capacitance Model Summary:

Gate-Channel Capacitance
- $C_{GC} \approx 0$ ($|V_{GS}| < |V_T|$)
- $C_{GC} = C_{ox}\cdot W\cdot L_{eff}$ (Linear)
  - 50% G to S, 50% G to D
- $C_{GC} = (2/3)\cdot C_{ox}\cdot W\cdot L_{eff}$ (Saturation)
  - 100% G to S

Gate Overlap Capacitance
- $C_{GSO} = C_{GDO} = C_O\cdot W$ (Always)

Junction/Diffusion Capacitance
- $C_{diff} = C_j\cdot L_S\cdot W + C_{jsw}\cdot(2L_S + W) + C_{jg}W$ (Always)

6 questions, pick 5; Read Them All First!
Take A Deep Breath! Stretch Your Hands!